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Design and Performance Evaluation of an Intelligent Adaptive Digital Communication Receiver using Automatic Modulation Classification

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CERTIFICATE

This is to certify that the Project Phase-II Report entitled "**Design and Performance Evaluation of an Intelligent Adaptive Digital Communication Receiver using Automatic Modulation Classification**" is presented & submitted by Hari Gupta, Virav Vanani, Surendra Gamit, bearing Roll No. U23EC166, U23EC182, U23EC195 of B.Tech. Final Year in the partial fulfillment of the requirement for the award of B.Tech. Degree in Electronics & Communication Engineering for academic year 2026-27.

They have successfully and satisfactorily completed their **Project Phase-II Evaluation** in all respects. We certify that the work is comprehensive, complete and fit for evaluation.

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Declaration

We hereby declare that the work presented in this report is original and has been carried out by us under the guidance of **Dr. Golak Santra**. The design, implementation, experiments, analysis, and documentation presented here are prepared for the Phase-II evaluation of the B.Tech. project. This report has not been submitted, either in part or in full, to any other university or institution for the award of any degree or diploma.

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Abstract

Automatic modulation classification is useful in receivers that must operate without complete prior knowledge of the transmitted waveform. The objective of this Phase-II project is to extend modulation recognition into a complete adaptive digital communication receiver. Instead of ending at a modulation label, the proposed system uses the estimated format to configure a demodulator and evaluates whether the transmitted information is recovered correctly.

The transmitter generates random information bits and maps them to BPSK, QPSK, 8-PSK, or 16-QAM symbols. The symbols pass through a block-Rayleigh fading and additive-white-Gaussian-noise channel with carrier phase and frequency offsets. At the receiver, pilot-assisted synchronization and frame-level channel equalization are applied. A penalized nearest-constellation estimator selects the modulation, after which the corresponding demodulator recovers the bits. The experiment records bit error rate, symbol error rate, modulation-decision accuracy, packet success rate, and effective throughput over an SNR range of -4 to 16 dB.

The final experiment uses 200 frames per SNR and a fixed seed of 42. At 16 dB, the receiver achieves 0.095% BER, 0.172% SER, 100% modulation-decision accuracy, 75.5% packet success, and 1.888 bits/symbol effective throughput. The results show that an AMC decision must be evaluated through the complete information path because a correct format decision and error-free bit recovery are different requirements. The report documents the Phase-II theory, communication architecture, algorithms, experimental setup, measured results, limitations, and future SDR extensions.

List of Abbreviations

AMC	Automatic Modulation Classification
AWGN	Additive White Gaussian Noise
BER	Bit Error Rate
CSI	Channel State Information
I/Q	In-phase and Quadrature
PHY	Physical Layer
QAM	Quadrature Amplitude Modulation
PSK	Phase Shift Keying
RF	Radio Frequency
SDR	Software Defined Radio
SER	Symbol Error Rate
SNR	Signal-to-Noise Ratio

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Chapter 1
Introduction

1.1 Motivation and Background

Modern wireless systems operate in congested and dynamic spectrum environments. The transmitter may change its modulation, the channel may introduce fading and phase rotation, and several users may share adjacent frequency bands. A conventional receiver is normally configured with the modulation and synchronization parameters before data reception begins. This assumption is unsuitable for a non-cooperative receiver, a cognitive-radio node, or a monitoring station that must process unknown signals.

Automatic modulation classification is the task of identifying the modulation format from received samples. It supports adaptive demodulation, spectrum monitoring, electronic support systems, interference analysis, and waveform-aware resource allocation. Classical approaches use likelihood tests, higher-order moments, cumulants, and cyclostationary features. These methods are physically interpretable, but their performance depends on the observation length, channel assumptions, and quality of synchronization [1, 2, 4].

Deep learning has provided another route by learning signal representations directly from I/Q samples. A neural classifier can identify local temporal patterns and phase relationships without requiring a separate handcrafted feature-design stage [7, 11]. However, a classification accuracy value alone does not demonstrate that a receiver can recover the transmitted information. The communication consequence of a wrong modulation decision must be measured after demodulation.

1.2 Applications of AMC

AMC is relevant to cognitive radio, where a receiver senses a waveform and selects a compatible processing chain. It is also useful in spectrum surveillance, where several unknown signals must be characterized from short observations. In adaptive wireless systems, modulation recognition can support mode selection, link adaptation, and interference-aware processing. In software-defined radio, the classifier can act as a control block between the digitized signal and a bank of candidate demodulators.

These applications impose requirements that are different from a static classification benchmark. The receiver must operate across channel conditions, report

confidence or uncertainty, and provide a measurable benefit to the recovered information. Phase II is designed around this system-level requirement.

1.3 Problem Definition and Objectives

Problem Statement: Design and evaluate an adaptive digital communication receiver that estimates the transmitted modulation and uses that estimate to recover information bits under noise, fading, phase offset, and frequency offset.

The objectives of the project are:

1. Generate reproducible information frames using BPSK, QPSK, 8-PSK, and 16-QAM.
2. Model a block-Rayleigh and AWGN channel with carrier impairments.
3. Apply pilot-assisted synchronization and channel equalization.
4. Estimate the modulation format using received constellation structure.
5. Select a compatible demodulator automatically.
6. Measure BER, SER, packet success, throughput, and decision accuracy.
7. Compare adaptive processing with a fixed-QPSK receiver.
8. Document the Phase-II implementation and experimental results clearly.

1.4 Contributions

The main contribution is a complete transmitter–channel–receiver simulation in which modulation recognition controls demodulation. The work adds four digital modulation formats, configurable channel impairments, frame-level correction, an interpretable estimator, adaptive bit recovery, and a fixed-modulation reference. It also introduces communication-level evaluation rather than reporting only a class label.

1.5 Scope and Limitations

The current work is a controlled software simulation. It includes a flat block-fading model, AWGN, carrier phase and frequency offsets, and pilot-assisted correction.

Pulse shaping, matched filtering, timing-loop implementation, forward-error correction, interleaving, retransmission, and over-the-air SDR measurements are outside the present scope. These limitations are stated so that the simulation results are not confused with hardware measurements.

1.6 Organization of the Report

Chapter 2 reviews classical AMC, deep-learning approaches, communication receiver design, and the research gap. Chapter 3 presents the mathematical background of complex baseband signals, digital modulation, channel impairments, synchronization, and evaluation metrics. Chapter 4 describes the generated Phase-II data, modulation classes, preprocessing, and experimental record. Chapter 5 explains the proposed receiver methodology and design. Chapter 6 describes implementation and experimental setup. Chapter 7 presents measured results and discussion. Chapter 8 concludes the work and identifies future extensions.

Chapter 2
Literature Review

2.1 Classical AMC Approaches

Early AMC systems used statistical moments, higher-order cumulants, likelihood functions, and cyclostationary features. Higher-order statistics can distinguish modulation families because the distribution of a received signal changes with the waveform. Cyclostationary methods exploit periodic features introduced by symbol timing, carrier structure, or framing [4, 5].

The advantage of classical methods is interpretability. A designer can identify which feature is responsible for a decision and can often estimate the computational complexity analytically. Their limitation is sensitivity to observation length and model mismatch. Noise, fading, carrier offset, frequency-selective propagation, and nonlinear hardware can alter the statistics on which the decision depends.

2.2 Likelihood and Feature-Based Methods

Likelihood-based AMC compares the received sequence with a probability model for every candidate modulation. When the channel and noise parameters are known, such methods can be statistically efficient. In practical systems, the required likelihood model is difficult to maintain because the receiver may not know the exact fading, synchronization, or hardware condition.

Feature-based systems reduce the computational burden by extracting moments, spectral features, cyclic features, or instantaneous amplitude and phase statistics. They can provide a useful compromise between complexity and performance, but their success depends on choosing features that remain discriminative at low SNR. Phase II uses a constellation-based estimator because it is transparent and directly connected to demodulator selection.

2.3 Deep Learning for AMC

Deep learning changed the AMC workflow by allowing features to be learned from raw I/Q sequences. Convolutional neural networks are attractive because one-dimensional filters can detect local waveform patterns, short phase transitions, and amplitude relationships. RadioML-style datasets have provided a common basis for studying these models [7, 8].

The main limitation of a classifier-only evaluation is that the output is often treated as the final result. A high classification accuracy does not directly specify the BER after a receiver uses the decision. Conversely, an incorrect decision may select an incompatible demodulator and cause a much larger information loss than a single classification error suggests.

2.4 Adaptive Receiver Design

An adaptive receiver contains a sensing or recognition stage followed by a configuration stage. The configuration stage may select a demodulator, symbol mapping, coding mode, timing method, or channel-estimation strategy. This makes receiver performance dependent on both recognition and subsequent signal processing.

The Phase-II design follows this principle. The estimator does not operate as an isolated classifier. Its output selects one of four demodulators, and the receiver compares the recovered bits with the transmitted bits. This creates a traceable relationship between waveform uncertainty and communication reliability.

2.5 Over-the-Air Generalization

Synthetic experiments provide repeatability and controlled comparison, but they do not include every effect present in an RF front-end. Oscillator phase noise, automatic-gain-control transients, antenna response, analog filtering, frequency-selective multipath, and timing drift can reduce performance when a system moves to an SDR. Over-the-air studies have shown the importance of these effects [8].

The present report therefore treats the simulation as an engineering validation stage. The channel model is sufficiently rich to expose modulation-decision and bit-recovery failures, while remaining reproducible for a project evaluation. Hardware validation is proposed as the next stage.

2.6 Benchmark Datasets and Tooling

RadioML 2016.10a is a widely used benchmark for raw-I/Q modulation classification. It is useful for retaining an earlier AMC baseline, but the new Phase-II link is evaluated with generated information frames so that BER and packet success can be measured directly. Python, NumPy, Matplotlib, and PyTorch provide the numerical and visualization environment used by the project [14].

2.7 Summary of Representative Studies

Table 2.1: Representative studies and their relation to the Phase-II work.

Reference	**Approach**	**Relevance to this project**
Gardner [5]	Cyclostationary analysis	Provides interpretable signal features for detection.
Azzouz and Nandi [4]	Statistical AMC	Shows the importance of feature and channel assumptions.
O'Shea and Corgan [7]	CNN on raw I/Q	Motivates data-driven waveform recognition.
O'Shea *et al.* [8]	Over-the-air classification	Demonstrates the need for realistic validation.
This project	Adaptive communication receiver	Connects modulation estimation to demodulation and BER.

2.8 Research Gap

Many AMC studies end after predicting a modulation label, while many link simulations assume that the modulation is already known. The engineering gap between these two stages is whether the recognition result can configure a useful receiver and deliver reliable information. Phase II addresses this gap by evaluating the complete path from generated bits to recovered bits.

Chapter 3
Theoretical Background

3.1 Baseband Signal Model

A passband signal can be represented in complex baseband form as

$$x[n] = I[n] + jQ[n], \tag{3.1}$$

where $I[n]$ and $Q[n]$ are the in-phase and quadrature components. The complex representation preserves amplitude and phase information while allowing the transmitter, channel, and receiver to be implemented as discrete-time operations.

For a frame of N symbols, the transmitter generates $x[0], x[1], \ldots, x[N-1]$. The receiver observes a corrupted sequence and must estimate the format before applying a symbol-to-bit mapping. The quality of this estimate affects every subsequent stage.

3.2 Digital Modulation Families

BPSK uses two opposite phase points and carries one bit per symbol. Its large decision separation makes it robust, but its spectral efficiency is low. QPSK uses four phase points and carries two bits per symbol, providing a common balance between robustness and rate. 8-PSK carries three bits per symbol by reducing angular separation. 16-QAM uses both amplitude and phase and carries four bits per symbol, but requires more accurate gain, phase, and noise control.

Table 3.1: Digital modulation formats used by the Phase-II receiver.

Format	**Order**	**Bits/symbol**	**Characteristic**
BPSK	2	1	Two phase states and large decision margin.
QPSK	4	2	Four phase states and moderate spectral efficiency.
8-PSK	8	3	Higher rate with reduced angular separation.
16-QAM	16	4	Amplitude and phase states for high rate.

3.3 Channel Impairments

The Phase-II received signal is modelled as

$$y[n] = h\, x[n] e^{j(2\pi \Delta f n + \phi)} + w[n], \tag{3.2}$$

where h is a complex block-Rayleigh channel coefficient, Δf is a normalized frequency offset, ϕ is a carrier phase offset, and $w[n]$ is complex AWGN. The coefficient is constant during one frame and changes between frames, allowing fading to affect packet-level performance.

Rayleigh fading changes the amplitude and phase of the signal without a dominant line-of-sight component. AWGN introduces random perturbations whose variance is controlled by the selected SNR. Phase offset rotates the entire constellation, while frequency offset produces a rotation that changes with sample index. These impairments test different aspects of receiver correction and constellation recognition.

3.4 Signal-to-Noise Ratio

The signal-to-noise ratio is defined as

$$\mathrm{SNR}_{dB} = 10 \log_{10} \left(\frac{P_s}{P_n} \right), \tag{3.3}$$

where P_s is the average signal power and P_n is the average noise power. The simulation normalizes the candidate constellations and scales complex Gaussian noise from the selected SNR so that the modulation formats are compared under one common convention.

3.5 Synchronization and Equalization

In a practical receiver, synchronization includes automatic gain control, timing recovery, carrier-frequency estimation, carrier-phase recovery, and channel estimation. Phase II uses a pilot-assisted simulation estimate for the frame-level carrier and channel parameters. This isolates the new adaptive-modulation problem from an unfinished hardware synchronization loop.

After carrier correction, the frame is equalized using

$$\hat{x}[n] = \frac{y[n]}{h}. \tag{3.4}$$

The corrected samples are normalized before their distance to each candidate constellation is calculated. The limitation of this controlled estimate is explicitly discussed in the results chapter.

3.6 AMC Approaches

The project distinguishes three receiver actions. A fixed receiver applies one known demodulator to every frame. A classifier-only receiver produces a label but does not recover information. The proposed adaptive receiver estimates the format and passes the decision to the corresponding demodulator. This third arrangement is the focus of Phase II.

3.7 Communication Metrics

The bit error rate and symbol error rate are

$$\mathrm{BER} = \frac{N_{\mathrm{bit,error}}}{N_{\mathrm{bit,total}}}, \qquad \mathrm{SER} = \frac{N_{\mathrm{symbol,error}}}{N_{\mathrm{symbol,total}}}. \tag{3.5}$$

A packet is successful only when every bit in its frame is recovered correctly. If b_m is the bits-per-symbol value of the selected modulation, the effective throughput is the average of b_m multiplied by the packet-success indicator. Therefore, packet success is stricter than BER and exposes the practical effect of even a small number of bit errors.

Chapter 4
Dataset and Preprocessing

4.1 Phase-II Data Generation

Unlike a classification-only experiment, the Phase-II link begins with information bits rather than pre-labelled signal windows. For every frame, a reproducible random generator creates a binary sequence. One of the four candidate modulation formats is selected, and the bits are grouped according to its order. The resulting complex symbols are retained as the transmitter reference for later error calculation.

Each frame contains 256 symbols. Thus, BPSK frames contain 256 information bits, QPSK frames contain 512 bits, 8-PSK frames contain 768 bits, and 16-QAM frames contain 1024 bits. Using the same symbol-frame length makes packet-level comparisons consistent while allowing the information rate to change with modulation order.

4.2 Modulation Classes

Table 4.1: Modulation classes used to generate Phase-II communication frames.

Class	**Bits/symbol**	**Use in the experiment**
BPSK	1	Robust low-order reference.
QPSK	2	Fixed-receiver comparison format.
8-PSK	3	Higher-order phase modulation.
16-QAM	4	High spectral-efficiency format.

4.3 Constellation Normalization

The constellation points are normalized to a common average-power convention. This avoids giving one format an artificial advantage because of a larger absolute amplitude. After channel correction, the received symbols are placed on the same scale before nearest-point distances are compared.

4.4 Channel Data Preparation

For each generated frame, the channel draws a block-Rayleigh coefficient, carrier phase, frequency offset, and complex AWGN realization. The SNR is selected from the sweep

$$-4, -2, 0, 2, 4, 6, 8, 10, 12, 14, 16 \text{ dB}.$$

The transmitted bits, transmitted symbols, received samples, estimated format, recovered bits, and error counts are retained until all frame-level metrics have been calculated.

4.5 Preprocessing at the Receiver

The receiver applies the inverse simulated carrier rotation and equalizes the block channel before estimation. The corrected samples are normalized using the candidate constellation scale. This operation is not a replacement for a practical timing loop; it is a controlled preprocessing stage that makes the contribution of adaptive modulation measurable.

4.6 Experiment Split and Repetition

The communication experiment is not divided into train, validation, and test subsets because it is a link simulation rather than a supervised training run. Instead, every SNR value is evaluated using 200 independently generated frames with seed 42. The same modulation list, frame length, channel model, and metric definitions are used at every SNR.

4.7 Data Integrity and Reproducibility

The complete numerical arrays are written to a JSON record. This record contains the SNR values, adaptive and fixed-receiver BER and SER, decision accuracy, packet success, throughput, modulation list, frame count, and seed. Figures are generated from the same record, preventing manual transcription errors between the experiment and the report.

4.8 Limitations of the Generated Data

The data is generated by a controlled software channel and therefore does not include antenna response, analog filtering, oscillator phase noise, frequency-selective

multipath, timing drift, or hardware nonlinearity. The generated frames are appropriate for algorithm verification and comparative analysis, but a future SDR experiment is required before making claims about over-the-air performance.

Chapter 5
Proposed Methodology and Model Design

5.1 End-to-End Pipeline

The proposed Phase-II pipeline is

$$\begin{aligned}\text{random bits} \rightarrow \text{modulator} \rightarrow \text{channel} \rightarrow \text{correction} \rightarrow \text{AMC decision} \rightarrow \\ \text{adaptive demodulator} \rightarrow \text{recovered bits.}\end{aligned} \tag{5.1}$$

The final bit sequence is compared with the transmitter reference. This pipeline is intentionally different from a classifier-only workflow because the modulation decision controls a subsequent physical-layer operation.

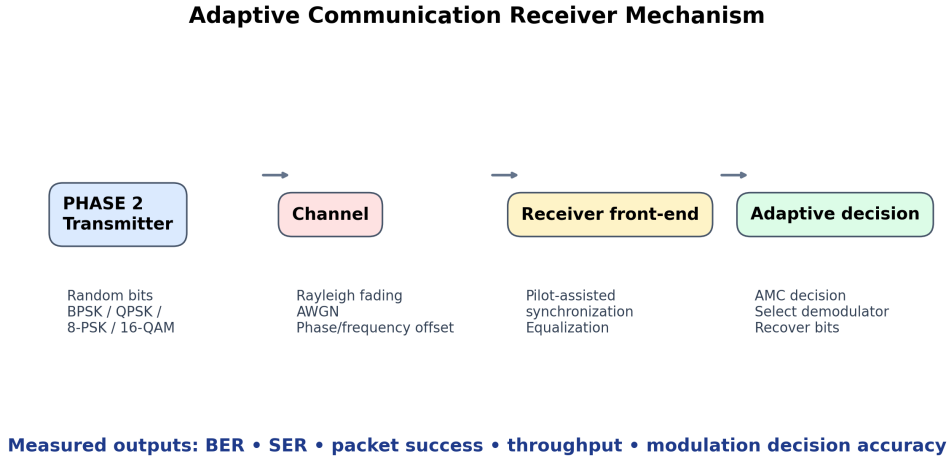


Figure 5.1: End-to-end Phase-II adaptive communication receiver.

5.2 Transmitter Model

The transmitter selects a modulation and groups the random bits into symbols. The mapper produces normalized complex points according to the selected constellation. The mapping is deterministic and is reused by the corresponding demodulator, so a clean round-trip test should recover all bits exactly.

5.3 Channel Model

The channel applies block fading, carrier rotation, and AWGN in sequence. Block fading changes the amplitude and phase of a complete frame. Carrier phase offset rotates the constellation, while frequency offset causes a progressive rotation. AWGN controls the SNR-dependent uncertainty that eventually produces both estimator errors and symbol errors.

5.4 Receiver Front-End

The receiver front-end reverses the simulated carrier rotation and divides by the block channel coefficient. The output is a corrected complex sequence. In an SDR implementation, this stage would be replaced by a timing recovery loop, pilot detector, carrier-recovery loop, and channel estimator. The current design keeps the estimate controlled so that the adaptive demodulation contribution can be isolated.

5.5 Constellation-Distance Estimator

Let $\mathcal{C}_m$ denote the normalized constellation for candidate format m. For each corrected received sample, the estimator finds the nearest point in $\mathcal{C}_m$ and calculates the average distance d_m. It then selects

$$\hat{m} = \arg\min_{m} \left(d_m + \lambda \log_2 |\mathcal{C}_m| \right), \tag{5.2}$$

where the penalty discourages a high-order constellation from winning only because it contains more possible points.

5.6 Adaptive Demodulator

The estimated format selects one of four demodulation rules. Each corrected sample is assigned to its nearest point, and the point index is converted back to bits using the transmitter's mapping. The recovered bits are aligned with the original frame and compared bit by bit. The estimator decision is recorded separately so that decision errors and within-format detection errors can be distinguished.

5.7 Fixed-QPSK Baseline

The fixed-QPSK receiver applies the QPSK demodulator regardless of the transmitted format. It represents a system that does not perform modulation adaptation. Its performance is expected to remain poor for BPSK, 8-PSK, and 16-QAM frames, but it provides a useful reference for measuring the value of adaptive selection.

5.8 Failure Propagation

The system has a clear failure chain. At low SNR, constellation points overlap and the estimator may choose the wrong format. The wrong format selects an incompatible demodulator, which can create many bit errors. Even when the estimated format is correct, noise can move individual points across decision boundaries. A single bit error is sufficient to mark a complete packet unsuccessful.

5.9 Design Rationale

The Phase-II estimator is deliberately interpretable. The candidate constellations, distance scores, selected modulation, recovered symbols, and final bit metrics can all be inspected. This design also provides a baseline for future replacement by a trained CNN or a confidence-aware estimator.

Chapter 6

Implementation and Experimental Setup

6.1 Software and Tools

The project is implemented in Python using NumPy for numerical operations, Matplotlib for figures, PyTorch for the retained learning components, and pytest for automated verification. The source is organized into reusable communication primitives, an experiment driver, visualization utilities, tests, and report artifacts. The report does not include executable command blocks; the repository documentation contains the operational instructions.

6.2 Experimental Environment

The final communication experiment is deterministic at the experiment level. The random seed is 42, the frame length is 256 symbols, and 200 frames are processed at each SNR. The channel and estimator parameters are held constant across the sweep, allowing changes in the curves to be interpreted primarily as SNR effects.

6.3 Hyperparameters and Link Parameters

Table 6.1: Phase-II link parameters used for the final experiment.

Parameter	**Value**
Modulation formats	BPSK, QPSK, 8-PSK, 16-QAM
Symbols per frame	256
SNR range	-4 to 16 dB
SNR increment	2 dB
Frames per SNR	200
Random seed	42
Channel	Block Rayleigh fading and AWGN
Carrier impairments	Phase and frequency offset
Correction	Pilot-assisted simulation estimate
Estimator	Penalized nearest-constellation distance

6.4 Evaluation Metrics

Five primary metrics are calculated. BER counts erroneous bits, SER counts erroneous symbols, and modulation-decision accuracy counts correct format estimates. Packet success requires every bit in a frame to be correct. Effective throughput weights the bits-per-symbol value of each successful frame by its packet-success indicator. These metrics are reported together because no single value captures the complete receiver behavior.

6.5 Experimental Procedure

At each SNR, the experiment generates 200 frames. For every frame it selects a modulation, creates bits, maps them to symbols, applies the channel, corrects the received sequence, estimates the modulation, demodulates the symbols, and calculates errors. The same transmitted frame is also processed by the fixed-QPSK reference. Aggregate arrays are saved before figures are created.

6.6 Verification

Automated tests cover clean modulation round trips, validation of invalid inputs, deterministic data handling, and clean constellation identification. The report source is checked for balanced LaTeX environments, existing figure references, and the required style dependencies. These checks supplement the numerical experiment.

6.7 Reproducibility Record

The JSON result record stores the seed, frame count, SNR list, modulation list, adaptive metrics, and fixed-QPSK metrics. The figures are generated from the same values used in the tables.

Chapter 7

Results and Discussion

7.1 Training Dynamics

Phase II is evaluated as a communication-link experiment rather than as a new supervised training run. Consequently, the primary result is the SNR-dependent receiver behavior, not a new training-loss curve. The retained recognition baseline motivates the AMC function, while the new evidence is the measured relationship between modulation selection, bit recovery, packet success, and throughput.

7.2 Overall Communication Performance

The adaptive receiver has two operating regions. From -4 to approximately 8 dB, noise and fading make the four constellations difficult to distinguish. From 10 dB onward, the modulation decision becomes reliable and the BER and SER decrease sharply. This transition is the most important Phase-II result because it exposes the operating limit of the complete communication chain.

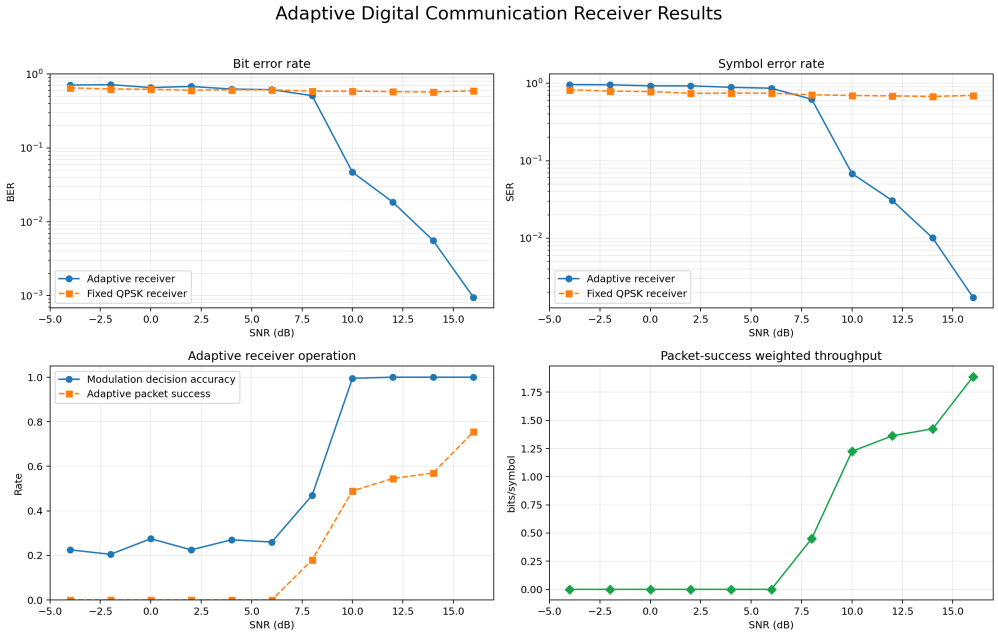


Figure 7.1:

BER, SER, decision accuracy, packet success, and throughput of the adaptive receiver.

7.3 SNR-Wise Performance

Table 7.1: Selected SNR-wise results from the final Phase-II experiment.

SNR	BER	SER	Decision	Packet	Throughput
10 dB	4.69%	6.94%	99.5%	49.0%	1.225
12 dB	1.85%	3.05%	100.0%	54.5%	1.363
14 dB	0.56%	1.00%	100.0%	57.0%	1.425
16 dB	0.095%	0.172%	100.0%	75.5%	1.888

At 10 dB, the decision accuracy is already 99.5%, but BER remains 4.69%. This demonstrates that correct modulation identification does not guarantee correct symbol decisions. At 12 dB, BER decreases to 1.85% and packet success increases to 54.5%. At 16 dB, the adaptive receiver enters its most reliable measured operating point.

7.4 High-SNR Performance

At 16 dB, measured BER is 0.095% and SER is 0.172%, with 100% modulation-decision accuracy. Packet success is 75.5% because a packet fails when any bit in its 256-symbol frame is incorrect. Effective throughput reaches 1.888 bits/symbol because successful frames contribute according to their modulation order.

7.5 Low-SNR Performance

At -4 dB, adaptive BER is approximately 0.710 and modulation-decision accuracy is 22.5%. Through 8 dB, the estimator remains uncertain and packet success is zero or low. This region is valuable for analysis because it reveals where an adaptive receiver needs a confidence threshold, fallback to a robust modulation, error-control coding, or retransmission.

7.6 Comparison with Fixed QPSK

The fixed-QPSK receiver maintains BER between approximately 0.57 and 0.65 over much of the sweep because it is mismatched to BPSK, 8-PSK, and 16-QAM frames. The adaptive receiver improves rapidly after the modulation decision transition and reaches 0.000948 BER at 16 dB. This comparison shows that the Phase-II contribution is receiver adaptation, not merely a change in how a label is reported.

7.7 Error Analysis and Observations

Three error sources dominate the results. First, noise causes constellation overlap and estimator ambiguity. Second, fading changes the received amplitude and effective SNR from frame to frame. Third, residual correction error moves points across decision boundaries even when the modulation decision is correct. Packet success combines all of these effects and is therefore stricter than BER or SER.

7.8 Practical Considerations

The current system is suitable for algorithm verification and controlled comparison. A real receiver must estimate channel and carrier parameters from noisy pilots, handle timing uncertainty, and process samples under latency and memory constraints. The estimator can later be replaced by a trained CNN, but the

communication metrics should remain because they measure the usefulness of the recognition decision.

7.9 Limitations of the Results

The experiment uses pilot-assisted simulation correction, flat block fading, an uncoded link, and a constellation-distance estimator. It does not include pulse shaping, timing recovery, frequency-selective multipath, Doppler, oscillator phase noise, nonlinear amplification, or over-the-air hardware. The numerical results should therefore be interpreted as a reproducible Phase-II software baseline.

Chapter 8
Conclusion and Future Work

8.1 Conclusion

This project developed a complete Phase-II adaptive digital communication receiver. The system generates information bits, supports four modulation formats, models channel and carrier impairments, corrects the received sequence, estimates the modulation, selects a demodulator, and evaluates recovered information through communication metrics.

The final experiment shows a clear SNR-dependent operating region. At 16 dB, the adaptive receiver achieves 0.095% BER, 0.172% SER, 100% decision accuracy, 75.5% packet success, and 1.888 bits/symbol throughput. The difference between 99.5% decision accuracy and 4.69% BER at 10 dB also demonstrates why end-to-end communication evaluation is necessary.

8.2 Contributions

- Designed a transmitter–channel–receiver Phase-II architecture.
- Implemented BPSK, QPSK, 8-PSK, and 16-QAM symbol processing.
- Added block-Rayleigh fading, AWGN, phase offset, and frequency offset.
- Implemented correction, constellation estimation, and adaptive demodulation.
- Added a fixed-QPSK reference for system-level comparison.
- Evaluated BER, SER, decision accuracy, packet success, and throughput.
- Created the result record, figures, tests, and supporting project files.

8.3 Limitations

The present receiver is a controlled simulation with pilot-assisted correction and no error-control coding. It does not yet include a complete timing loop, online CNN inference, multipath, Doppler, or SDR measurements. These limitations define the boundary of the current claims rather than invalidating the measured software comparison.

8.4 Future Work

The next stage should implement noisy-pilot timing and carrier recovery, replace the controlled correction with a practical channel estimator, and connect the learned classifier to the live receiver path. Confidence-aware fallback, coding, interleaving, retransmission, frequency-selective channels, Doppler, oscillator phase noise, and nonlinear RF effects should then be evaluated. Finally, the complete system should be tested on an SDR using recorded and over-the-air signals.

Appendix A
Modulation Class Descriptions

A.1 Summary of Modulation Characteristics

Table A.1: Modulation classes used by the Phase-II communication link.

Class	Bits/symbol	Description
BPSK	1	Two opposite phase states; robust and low-rate.
QPSK	2	Four phase states; balanced robustness and rate.
8-PSK	3	Eight phase states; higher rate and smaller phase separation.
16-QAM	4	Amplitude and phase states; high rate and higher sensitivity.

A.2 Communication-Metric Definitions

Metric	Meaning in this project
BER	Fraction of recovered bits that differ from transmitted bits.
SER	Fraction of recovered symbols that differ from transmitted symbols.
Decision accuracy	Fraction of frames with the correct modulation estimate.
Packet success	Fraction of frames with every bit recovered correctly.
Throughput	Successful information bits per transmitted symbol.

Appendix B
Project Workflow Summary

B.1 Phase-II Workflow

The project workflow consists of system definition, constellation construction, channel modelling, receiver correction, modulation estimation, adaptive demodulation, metric calculation, and result interpretation. Each stage is separated so that the receiver can later be connected to an SDR front-end or a learned online classifier.

1. Define the four modulation formats and normalized constellations.
2. Generate random bits and map them to symbols.
3. Apply block-Rayleigh fading, AWGN, phase offset, and frequency offset.
4. Correct the simulated carrier and channel effects.
5. Estimate the modulation using penalized constellation distance.
6. Demodulate using the selected format and recover the bits.
7. Calculate BER, SER, packet success, throughput, and decision accuracy.
8. Store the complete numerical record and generate report figures.
9. Verify the implementation with automated tests and package checks.

B.2 Reproducibility Record

The fixed seed is 42, the frame length is 256 symbols, the SNR values run from -4 to 16 dB in 2 dB steps, and 200 frames are evaluated at every point. The JSON result record contains the complete arrays used in the figures. The archive also contains the report source, style dependency, architecture figure, result figure, communication module, experiment driver, and tests.

B.3 Closing Note

The report follows the same preliminary pages, chapter sequence, appendix format, and reference-page arrangement as the earlier project report. The content in

Chapters 1–8 describes the Phase-II communication receiver, its implementation, and its measured results.

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